Plant leaf disease detection on Edge Devices.

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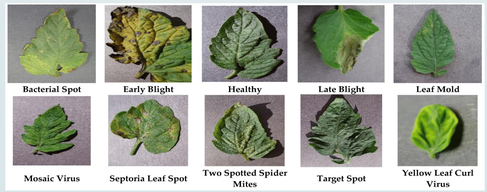
Abstract

This thesis focuses on exploring and implementing techniques in edge computing for the task of plant leaf disease identification. The aim is to run object classification model inference on memory constrained devices such as microcontrollers and single board computers in an optimum way to get a high speed inference with good accuracy and less power consumption. It talks about various limitations and trade-offs these devices has to undergo. Further this thesis also explores ways to optimize the machine learning model through techniques like pruning and quantization in order to make the models edge hardware friendly.

Introduction

- This work tries to use Convolutional neural networks (CNN) for identifying the disease features in the leaves.



- This work explores the usage of

In this work we will 2 edge devices, to run the ML model inferences on them

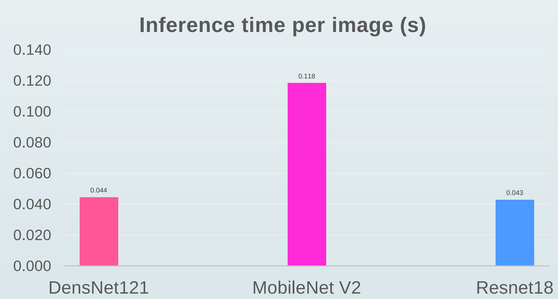
- Jetson Nano by NVIDIA
- STM32 microcontroller by STMicroelectronics.

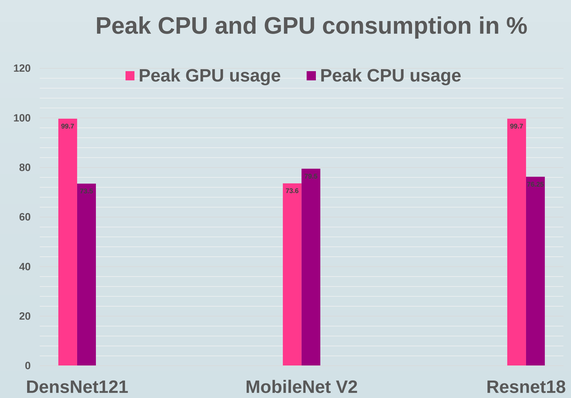
- Models under consideration.
- ResNets
- DenseNet
- MobileNet.
- Custom models derived from above models.

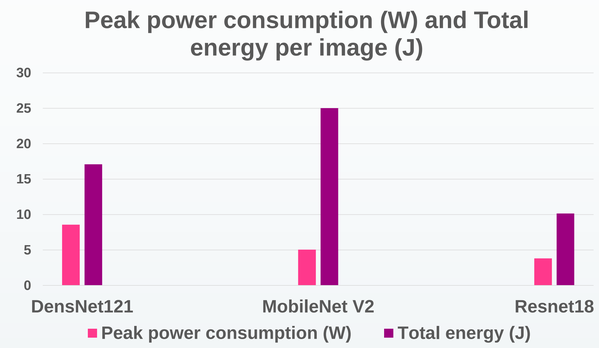
Methodology for using Jetson Nano

- Jetson Nano is a single board computer with
  - -Quad core ARM A57 processor
  - -4 GB RAM
  - -128 core Maxwell GPU
  - -Jetpack SDK for software development.
- Inputs processed in batches of 32.
- Models pretrained on ImageNet dataset
- Key performance parameters of interest:
  - -RAM usage
  - -GPU usage
  - -CPU usage
  - -Power and Energy consumption
  - -Inference time

Results with Jetson Nano



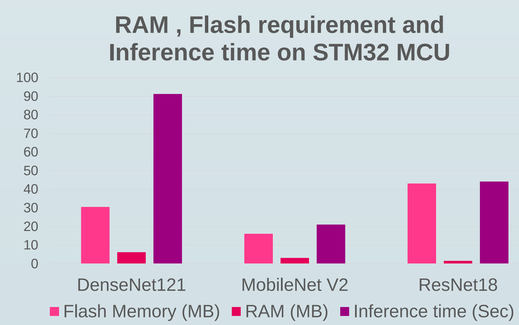




- Advantages:
- Real time performance
- OS environment, use of high level languages.
- Disadvantages:
- High cost, around INR 20-25k
- High power consumption.

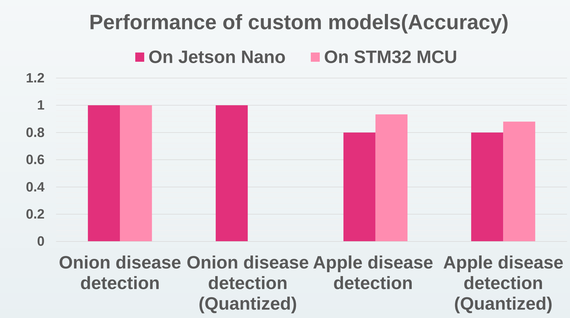
STM32 Microcontroller

- STM32 microcontrollers(MCU) STM32F769DISC has
  - Single core ARM Cortex M7 processor.
  - Compact and cheaper solution.
  - 2MB Flash, 512KB RAM, 216MHz clock freq.
  - External memories for bigger models.
- High inference times.
- Peak power consumption of 1.65W
- No OS, no scope for running high level abstraction codes.

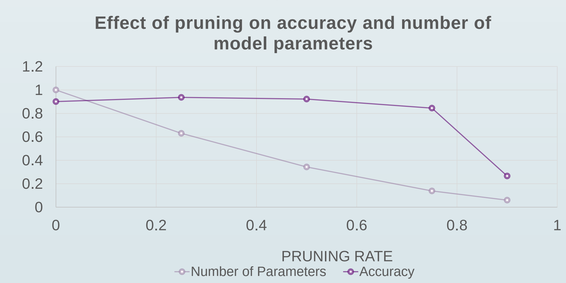


Custom made models for disease detection

- Models developed for
  - Detecting onion disease (Binary)
  - Detecting apple diseases (4 class)
  - Onion model used DenseNet121 backbone.
  - Apple model is smaller model with 4 conv layers.
  - Models also quantized to INT8 datatype.



- Development of filter pruning algorithm using filter norm
- Up to 86.4% reduction in size achieved with just ~6% loss in accuracy



Future work

- This work will extend to build a complete system pipeline to capture, preprocess, and infere the image with minimum resource consumption and maximum throughput

- Implementation of federated learning architecture to train local models on individual edge devices, and send the update on master model on the server.